

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

SPARSE CODING

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Dictionaries

Goal: find a **sparse representation** for our data.

- Data point $x \in \mathbb{R}^d$
- Sparse representation $z \in \mathbb{R}^m$
- Want: $x \approx \Phi z$, for some $d \times m$ matrix Φ

Think of the columns of Φ as **dictionary elements**:

$$\begin{pmatrix} \uparrow \\ x \\ \downarrow \end{pmatrix} \approx \underbrace{\begin{pmatrix} \uparrow & \uparrow & \cdots & \uparrow \\ \phi_1 & \phi_2 & \cdots & \phi_m \\ \downarrow & \downarrow & \cdots & \downarrow \end{pmatrix}}_{\text{dictionary } \Phi} \begin{pmatrix} \uparrow \\ z \\ \downarrow \end{pmatrix}$$

Sparse Coding

Given data points $x^{(1)}, \dots, x^{(n)} \in \mathbb{R}^d$, and an integer m .

- Choose m dictionary vectors $\phi_1, \dots, \phi_m \in \mathbb{R}^d$.
- Approximate each $x^{(i)}$ by a linear combination of these dictionary elements.

$$\underbrace{\begin{pmatrix} \uparrow & & \uparrow \\ x^{(1)} & \dots & x^{(n)} \\ \downarrow & & \downarrow \end{pmatrix}}_{\text{data matrix } X} \approx \underbrace{\begin{pmatrix} \uparrow & \uparrow & & \uparrow \\ \phi_1 & \phi_2 & \dots & \phi_m \\ \downarrow & \downarrow & & \downarrow \end{pmatrix}}_{\text{dictionary } \Phi} \underbrace{\begin{pmatrix} \uparrow & & \uparrow \\ z^{(1)} & \dots & z^{(n)} \\ \downarrow & & \downarrow \end{pmatrix}}_{\text{encoding } Z}$$

We want the columns of Z to be **sparse** (i.e., mostly zero).

Express $X \approx \Phi Z$: this is **matrix factorization**. How to find Φ, Z ?

Learning the Dictionary for Sparse Coding

Given $x^{(1)}, \dots, x^{(n)} \in \mathbb{R}^d$, find dictionary vectors $\phi_1, \dots, \phi_m$ and sparse representations $z^{(1)}, \dots, z^{(n)} \in \mathbb{R}^m$ such that

$$x^{(i)} \approx \Phi z^{(i)}.$$

Optimization problem: find matrices Φ, Z that minimize

$$\begin{aligned} L(\Phi, Z) &= \|X - \Phi Z\|_F^2 - \lambda \cdot \text{sparsity}(Z) \\ &= \sum_{i=1}^n \left(\|x^{(i)} - \Phi z^{(i)}\|^2 - \lambda \cdot \text{sparsity}(z^{(i)}) \right) \end{aligned}$$

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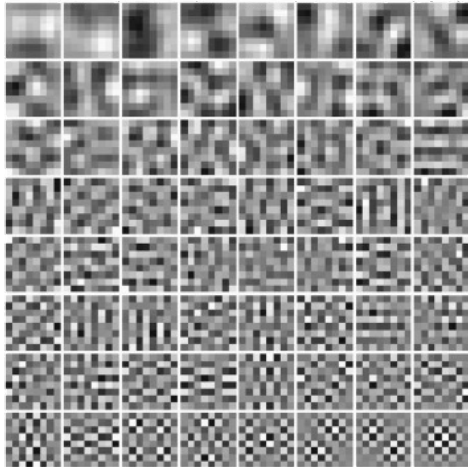
Learning Φ, Z by **alternating minimization**:

- Initialize Φ somehow
- Repeat until convergence:
 - Fixing Φ , minimize $L(\cdot)$ over Z
 - Fixing Z , minimize $L(\cdot)$ over Φ

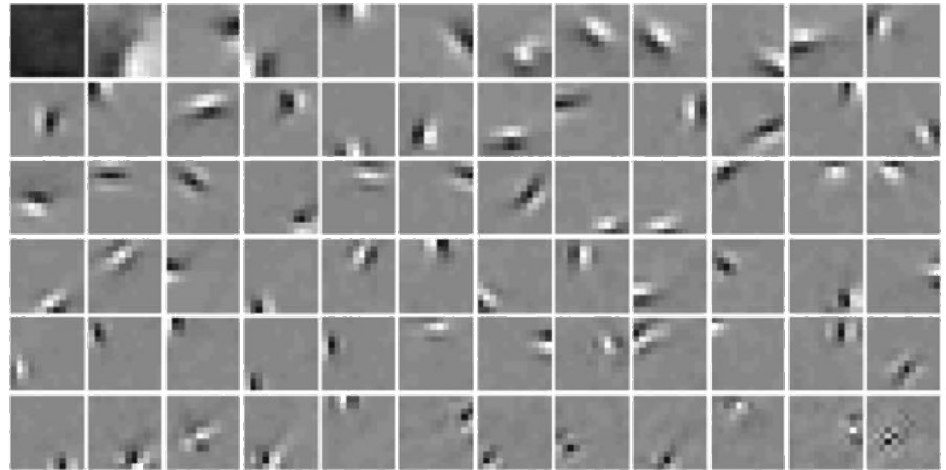
Example: Image Patches

Olshausen-Field (1996), Lewicki-Olshausen (1999): PCA versus sparse coding for natural image patches.

PCA

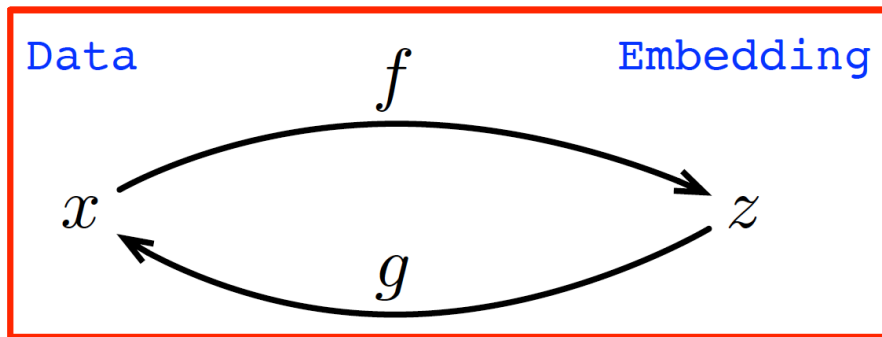


sparse coding



Sparse coding does a much better job at finding a basis that resembles the receptive fields of simple cells in visual cortex.

Back to the Autoencoder Formalism



- f : encoding map
- g : decoding map

How does sparse coding fit into this framework?

Neural Net-Based Sparse Autoencoders

